

DES Pune University, Pune
School of Engineering and Technology
Department of Computer Engineering and Technology
Program: B. Tech in Computer Science and Engineering

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Subject: Python Programming

Assignment No.: 12

Date: 12/01/2026

Lab Assignment: 12

Aim:

Write a program to sum the series $-1 + 1/2 + \dots + 1/n$.

Objective:

- To understand the use of loop for series calculation.
- To perform division and addition operations in Python.
- To calculate the sum of a mathematical series.

Theory:

A series is a sequence of numbers added together.

In this program, the first term is -1 and the remaining terms are fractions from $1/2$ up to $1/n$.

A loop is used to add each term of the series one by one and store the result in a variable.

Algorithm:

- Start
- Read the value of n
- Initialize `sum = -1`
- Print -1
- Repeat for i from 2 to n
 - Print $+ 1/i$
 - Add $1/i$ to `sum`
- Print the final value of `sum`
- Stop

Pseudo Code:

```
START
READ n
sum = -1
PRINT "-1"
FOR i = 2 TO n
    PRINT "+ 1/i"
```

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```
sum = sum + (1 / i)
END FOR
PRINT sum
STOP
```

Code:

```
n = int(input("Enter the value of n: "));
sum = -1;
print("-1", end="");
for i in range(2, n + 1):
    print(" + 1/" + str(i), end="");
    sum = sum + (1 / i);
print();
print("Sum of the series is:", sum);
```

Output:

```
Enter the value of n: 5
-1 + 1/2 + 1/3 + 1/4 + 1/5
Sum of the series is: 0.2833333333333333
```

Conclusion:

The program correctly calculates the sum of the series $-1 + 1/2 + 1/3 + \dots + 1/n$ using a simple loop.